Normalization (Up to 3NF or BCNF)

Table 1: User

1. Functional Dependencies (FDs)

- **User_ID** → Name, Email, Password, Role, Phone, Created_Date
- **Email** → User_ID, Name, Password, Role, Phone, Created_Date
- **Role_ID** → Role_Name (after decomposition)

Assumptions:

- User_ID is the Primary Key.
- Email is unique and can serve as a candidate key.
- Role values repeat, suggesting the need for a separate Role table.

2. Normal Form Identification

- **1NF:** Satisfied (atomic values, no repeating groups).
- **2NF:** Satisfied (all non-key attributes depend fully on the primary key).
- **3NF:** Not satisfied completely due to potential transitive dependency via Email and unnormalized Role values.

3. Decomposition to 3NF/BCNF

Table1: Role

Role_ID(PK)

Role_Name

Table2: User

User_ID (PK)

Email (Unique)

Password

Role_ID (FK)

Created_Date

Table3: User_Profile

User_ID(FK)
Name
Phone

4. Final Normalized Schema (3NF/BCNF)

Role

Role_ID	Role_Name

User

User_ID	Email	Password	Role_ID	Created_Date

User_Profile

User_ID	Name	Phone

Table 2: Space

1. **Functional Dependencies (FDs)**
   - Space_ID → Name, Location, Description, Capacity, Noise_Level, Opening_Hours, Created_Date
2. **Normal Form Identification**
   - 1NF: All columns contain single, simple values. No repeating groups.
   - 2NF: All non-key columns depend entirely on the primary key (Space_ID).
   - 3NF: No transitive dependencies exist.
   - BCNF: Every determinant is a candidate key.
3. **Decomposition to 3NF/BCNF**

   The table is in BCNF. No decomposition is required.

4. **Final Normalized Schema (3NF/BCNF)**

Space

Space_ID	Name	Location	Description	Capacity	Noise _Level	Open_hours	Created_date

Table 3: Amenities

1. **Functional Dependencies (FDs)**
   - Amenity_ID -> Name, Description
   - Name -> Amenity_ID, Description

   Candidate Keys:

   - Amenity_ID (Primary Key)
   - Name (Alternate Key, assuming Name is unique across all amenities)

2. **Normal Form Identification**

- Functional Dependencies (FDs)
- Amenity_ID -> Name, Description
- Name -> Amenity_ID, Description

3. **Decomposition to 3NF/BCNF**

   The table is in BCNF, provided Name is guaranteed unique. No decomposition is required.

4. **Final Normalized Schema(3NF/BCNF)**

Amenities

Amenity_ID	**Name**	**Description**

Table 4: Space_Amenities(Junction Table)

1. **Functional Dependencies (FDs)**

- No non-trivial functional dependencies (only attributes are the composite primary key).

Candidate Key:

- (Space_ID, Amenity_ID) - Composite Primary Key

2. Normal Form Identification
   - 1NF: All attributes are atomic.
   - 2NF: No non-key attributes to have partial dependencies.
   - 3NF: No transitive dependencies possible.
   - BCNF: No determinants other than candidate keys.

3. **Decomposition to 3NF/BCNF**
   - The table is in BCNF. No decomposition is required.

4. **Final Normalized Schema(3NF/BCNF)**

Amenities_space

Space_ID	Amenity_ID

Table 5: Bookings

1. **Functional Dependencies (FDs)**
   - Booking_ID -> User_ID, Space_ID, Booking_Date, Start_Time, End_Time, Status, Created_Date

Candidate Key:

- Booking_ID (Primary Key)

2. **Normal Form Identification**
   - 1NF: All attributes are atomic.
   - 2NF: No partial dependencies.
   - 3NF: No transitive dependencies.
   - BCNF: Every determinant is a candidate key.
3. **Decomposition to 3NF/BCNF**
   - The table is in BCNF. No decomposition is required.
4. **Final Normalized Schema(3NF/BCNF)**

Booking

Booking_ID	User_ID	Space_ID	Booking_Date	Start_Time	End_Time	Status	Created_Date

Table 6: Review

1. **Functional Dependencies (FDs)**
   - Review_ID -> User_ID, Space_ID, Rating, Comment, Review_Date
   - (User_ID, Space_ID) -> Review_ID, Rating, Comment, Review_Date

   Candidate Keys:

   - Review_ID (Primary Key — surrogate system-generated key)
   - (User_ID, Space_ID) - Alternate Key (business rule: one review per user per space)
2. **Normal Form Identification**
   - All non-key attributes depend fully on both candidate keys.
   - There are no partial dependencies.
   - There are no transitive dependencies.
   - All determinants (Review_ID, (User_ID, Space_ID)) are candidate keys.
3. **Decomposition to 3NF/BCNF**
   - The table is in BCNF. No decomposition is required.
4. **Final Normalized Schema(3NF/BCNF)**

Review

Review_ID	User_ID	Space_ID	Rating	Comment	Review_Date